PRODUCT REQUIREMENTS DOCUMENT

# Financial Fraud Detector

An AI-Assisted Financial Statement Anomaly Detection System

```
<div class="cover-meta">
  <table>
    <tr><td>Project Name</td><td>Financial Fraud Detector</td></tr>
    <tr><td>Version</td><td>1.0 (MVP)</td></tr>
    <tr><td>Document Type</td><td>Product Requirements Document (PRD)</td></tr>
    <tr><td>Date</td><td>2026</td></tr>
    <tr><td>Status</td><td>Ready for Development</td></tr>
    <tr><td>Target Platform</td><td>Web Application (Full Stack)</td></tr>
  </table>
</div>
```

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# 1. Executive Summary

The Financial Fraud Detector is a full-stack web application that analyzes a company's historical financial statements to detect anomalies that may indicate accounting irregularities, financial manipulation, or elevated financial reporting risk.
The system:
Accepts Excel/CSV financial data as input
Extracts raw financial numbers
Calculates financial ratios
Detects 7 types of anomalies across multiple years
Assigns a Financial Risk Score (0–100)
Uses AI to generate plain-English explanations
Displays everything on an interactive dashboard
⚠ **Important:** The system does NOT claim fraud. It identifies patterns that warrant further investigation.

# 2. Problem Statement

Financial statements contain large amounts of data, making it difficult to manually identify subtle relationships and anomalies across several years. Traditional analysis focuses on individual ratios, but fraud becomes visible when multiple financial variables are analyzed together over time.

### Example:

Revenue ↑ 10%
Accounts Receivable ↑ 45%

Operating Cash Flow ↓ 20%
Individually, these may not indicate a problem. Together, they warrant investigation.

# 3. Objectives

Accept historical financial statements (Excel/CSV)
Extract important financial data
Calculate relevant financial ratios
Compare metrics across multiple years
Detect 7 types of anomalies
Assign a Financial Risk Score (0–100)
Generate AI explanations for each anomaly
Display findings on an interactive dashboard

# 4. Target Users

| User Type | Description | Primary Need |
|---|---|---|
| **Financial Analyst** | Professional analyzing company financials | Quick risk assessment |
| **Auditor** | Internal/external auditor | Identify red flags for investigation |
| **Student/Researcher** | Academic user studying fraud detection | Learn anomaly patterns |
| **Investor** | Individual analyzing potential investments | Due diligence screening |

# 5. User Stories

## 5.1 Upload & Analysis

| ID | User Story | Priority |
|---|---|---|
| US-01 | As a user, I want to upload an Excel/CSV file with financial data so that the system can analyze it | High |
| US-02 | As a user, I want to see a loading indicator while the system processes my file | High |
| US-03 | As a user, I want to see an error message if my file format is wrong | High |
| US-04 | As a user, I want to select from previously uploaded companies | Medium |

## 5.2 Dashboard & Results

| ID | User Story | Priority |
|---|---|---|
| US-05 | As a user, I want to see a Financial Risk Score (0–100) prominently displayed | High |
| US-06 | As a user, I want to see the Risk Level (Low/Moderate/High/Very High) | High |
| US-07 | As a user, I want to see a list of all detected anomalies | High |
| US-08 | As a user, I want to see how many risk points each anomaly contributed | High |
| US-09 | As a user, I want to see interactive charts of financial trends over time | High |
| US-10 | As a user, I want to see AI-generated explanations for each anomaly | High |
| US-11 | As a user, I want to see recommended investigation steps for each anomaly | Medium |

## 5.3 Navigation

| ID | User Story | Priority |
|---|---|---|
| US-12 | As a user, I want to navigate between dashboard, upload, and company list pages | High |
| US-13 | As a user, I want to go back to the dashboard from any page | Medium |

| ID | User Story | Priority |
|---|---|---|
| US-14 | As a user, I want to compare multiple companies side by side | Low (Advanced) |

# 6. Functional Requirements

## 6.1 Data Input

| ID | Requirement | Details |
|---|---|---|
| FR-01 | Accept Excel (.xlsx, .xls) files | Max size: 10 MB |
| FR-02 | Accept CSV (.csv) files | Max size: 10 MB |
| FR-03 | Parse financial data from uploaded file | Use pandas |
| FR-04 | Validate required columns exist | If missing, show error |
| FR-05 | Validate minimum 3 years of data | If less, show warning |
| FR-06 | Store extracted data in database | SQLite |
| FR-07 | Handle missing/null values gracefully | Skip or interpolate |

## 6.2 Data Extraction

| ID | Requirement | Source |
|---|---|---|
| FR-08 | Extract Revenue | Income Statement |
| FR-09 | Extract COGS | Income Statement |
| FR-10 | Extract Operating Expenses | Income Statement |
| FR-11 | Extract Net Profit | Income Statement |
| FR-12 | Extract Accounts Receivable | Balance Sheet |
| FR-13 | Extract Inventory | Balance Sheet |
| FR-14 | Extract Total Assets | Balance Sheet |
| FR-15 | Extract Total Debt | Balance Sheet |
| FR-16 | Extract Shareholders' Equity | Balance Sheet |
| FR-17 | Extract Operating Cash Flow | Cash Flow Statement |
| FR-18 | Extract Capital Expenditure (Capex) | Cash Flow Statement |
| FR-19 | Extract Depreciation | Cash Flow Statement |
| FR-20 | Extract Related-Party Transaction Amounts | Notes |

## 6.3 Ratio Calculation

| ID | Ratio | Formula |
|---|---|---|
| FR-21 | Gross Margin | (Revenue – COGS) ÷ Revenue |
| FR-22 | Operating Margin | Operating Profit ÷ Revenue |
| FR-23 | Net Profit Margin | Net Profit ÷ Revenue |
| FR-24 | ROE | Net Profit ÷ Shareholders' Equity |

| ID | Ratio | Formula |
|---|---|---|
| FR-25 | ROA | Net Profit ÷ Total Assets |
| FR-26 | ROCE | Operating Profit ÷ Capital Employed |
| FR-27 | Current Ratio | Current Assets ÷ Current Liabilities |
| FR-28 | Quick Ratio | (Current Assets − Inventory) ÷ Current Liabilities |
| FR-29 | Operating Cash Flow Ratio | Operating Cash Flow ÷ Current Liabilities |
| FR-30 | Debt-to-Equity | Total Debt ÷ Shareholders' Equity |
| FR-31 | Debt-to-Assets | Total Debt ÷ Total Assets |
| FR-32 | Interest Coverage | Operating Profit ÷ Interest Expense |
| FR-33 | Receivables Turnover | Revenue ÷ Accounts Receivable |
| FR-34 | Inventory Turnover | COGS ÷ Inventory |
| FR-35 | Asset Turnover | Revenue ÷ Total Assets |
| FR-36 | Cash Conversion Cycle | Days Inventory + Days Receivables − Days Payables |
| FR-37 | OCF / Net Profit | Operating Cash Flow ÷ Net Profit |
| FR-38 | Free Cash Flow | Operating Cash Flow − Capex |

## 6.4 Anomaly Detection (7 Rules)

| ID | Anomaly | Condition | Risk Points |
|---|---|---|---|
| FR-39 | A. Receivables vs Revenue | (Receivables Growth − Revenue Growth) > 20% | +15 |
| FR-40 | B. Profit vs Operating Cash Flow | (Net Profit Growth > 0) AND (OCF Growth < 0) | +20 |
| FR-41 | C. Inventory Anomaly | (Inventory Growth − Revenue Growth) > 25% | +10 |
| FR-42 | D. Margin Anomalies | Current Net Margin > (Historical Avg + 5%) | +8 |
| FR-43 | E. Related-Party Transactions | Related-Party Amount Growth > 50% YoY | +12 |
| FR-44 | F. Depreciation vs Capex | (Capex Growth > 30%) AND (Depreciation Growth < 5%) | +10 |
| FR-45 | G. Debt & Interest Anomalies | (Debt Growth > 30%) AND (Interest Growth < 10%) | +5 |

### Detailed Rules

#### Rule A: Receivables vs Revenue

```
For each year i (from year 2 onwards):
  revenue_growth = (Revenue[i] - Revenue[i-1]) / Revenue[i-1] * 100
  receivables_growth = (Receivables[i] - Receivables[i-1]) / Receivables[i-1] * 100
  difference = receivables_growth - revenue_growth

  IF difference > 20:
    Trigger anomaly
    Risk Points: +15
    Description: "Receivables grew {receivables_growth}% vs Revenue {revenue_growth}%"
```

#### Rule B: Profit vs Operating Cash Flow

```
For each year i (from year 2 onwards):
  profit_growth = (NetProfit[i] - NetProfit[i-1]) / NetProfit[i-1] * 100
  ocf_growth = (OCF[i] - OCF[i-1]) / OCF[i-1] * 100

  IF profit_growth > 0 AND ocf_growth < 0:
    Trigger anomaly
    Risk Points: +20
    Description: "Profit grew {profit_growth}% but OCF declined {ocf_growth}%"
```

**Rule C: Inventory Anomaly**

```
For each year i (from year 2 onwards):
  revenue_growth = (Revenue[i] - Revenue[i-1]) / Revenue[i-1] * 100
  inventory_growth = (Inventory[i] - Inventory[i-1]) / Inventory[i-1] * 100
  difference = inventory_growth - revenue_growth

  IF difference > 25:
    Trigger anomaly
    Risk Points: +10
    Description: "Inventory grew {inventory_growth}% vs Revenue {revenue_growth}%"
```

**Rule D: Margin Anomalies**

```
historical_avg_margin = average of Net Margin for years 1 to n-1
current_margin = Net Margin for year n

IF current_margin > (historical_avg_margin + 5):
  Trigger anomaly
  Risk Points: +8
  Description: "Net margin jumped from {historical_avg}% to {current}%"
```

**Rule E: Related-Party Transactions**

```
For each year i (from year 2 onwards):
  rpt_growth = (RPT[i] - RPT[i-1]) / RPT[i-1] * 100

  IF rpt_growth > 50:
    Trigger anomaly
    Risk Points: +12
    Description: "Related-party transactions grew {rpt_growth}%"
```

**Rule F: Depreciation vs Capex**

```
For each year i (from year 2 onwards):
  capex_growth = (Capex[i] - Capex[i-1]) / Capex[i-1] * 100
  dep_growth = (Depreciation[i] - Depreciation[i-1]) / Depreciation[i-1] * 100

  IF capex_growth > 30 AND dep_growth < 5:
    Trigger anomaly
    Risk Points: +10
    Description: "Capex grew {capex_growth}% but depreciation only grew {dep_growth}%"
```

**Rule G: Debt & Interest Anomalies**

```
For each year i (from year 2 onwards):
  debt_growth = (TotalDebt[i] - TotalDebt[i-1]) / TotalDebt[i-1] * 100
  interest_growth = (InterestExpense[i] - InterestExpense[i-1]) / InterestExpense[i-1] * 100

  IF debt_growth > 30 AND interest_growth < 10:
    Trigger anomaly
```

```
    Risk Points: +5
    Description: "Debt grew {debt_growth}% but interest only grew {interest_growth}%"
```

## 6.5 Risk Scoring

| ID | Requirement | Details |
|---|---|---|
| FR-46 | Sum all risk points from detected anomalies | Total = sum of all points |
| FR-47 | Cap score at 100 | min(total, 100) |
| FR-48 | Assign risk level based on score | See table below |

### Risk Level Mapping

| Score Range | Risk Level | Color |
|---|---|---|
| 0–20 | Low Risk | Green |
| 21–40 | Moderate-Low Risk | Yellow |
| 41–60 | Moderate Risk | Orange |
| 61–80 | High Risk | Red |
| 81–100 | Very High Risk | Dark Red |

## 6.6 AI Explanation

| ID | Requirement | Details |
|---|---|---|
| FR-49 | Generate explanation for each anomaly | Via LLM API |
| FR-50 | Explanation has 3 parts | Detected, Why It Matters, Recommended Investigation |
| FR-51 | Use simple language | No jargon |
| FR-52 | Never claim fraud | Use "may indicate", "could suggest" |
| FR-53 | Store explanations in database | For reuse |

### AI Prompt Template

```
You are a financial analyst explaining anomalies to a non-expert.

Anomaly Detected: {anomaly_description}
Company: {company_name}
Year: {year}
Risk Points: {risk_points}

Please provide:


DETECTED ANOMALY: What was found (1 sentence)

WHY IT MATTERS: What this could indicate (2-3 sentences)

RECOMMENDED INVESTIGATION: What to check next (2-3 bullet points)


Keep language simple. Do not claim fraud. Use words like "may indicate" or "could suggest."
```

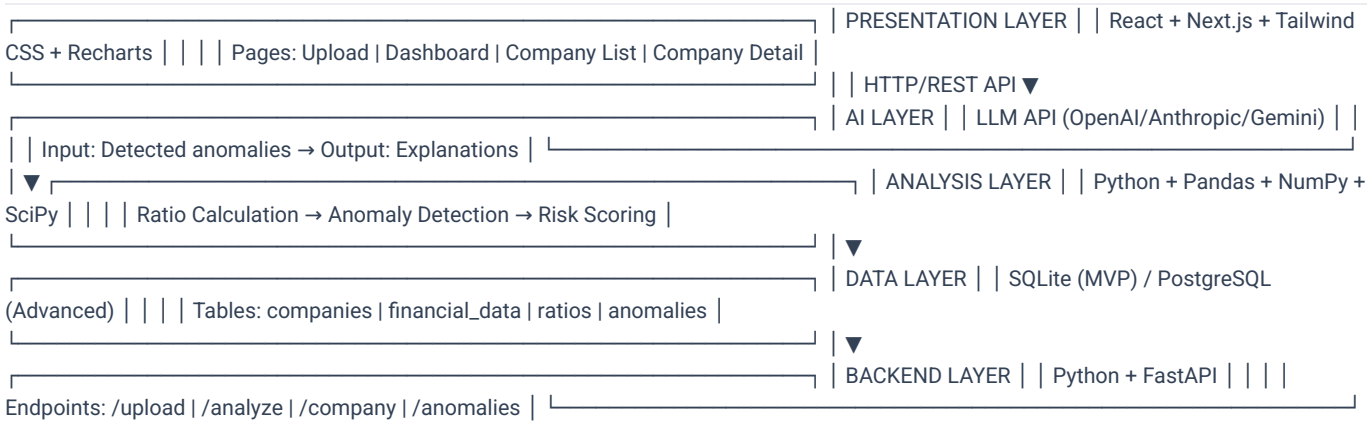
## 6.7 Dashboard Display

| ID | Requirement | Details |
|---|---|---|
| FR-54 | Display Risk Score prominently | Large number, 0–100 |
| FR-55 | Display Risk Level | Color-coded |
| FR-56 | Display list of anomalies | With risk points |
| FR-57 | Display interactive charts | Revenue, Profit, OCF, Receivables, Inventory, Debt, Capex, Depreciation |
| FR-58 | Display AI explanations | For each anomaly |
| FR-59 | Display company name | At top |
| FR-60 | Allow navigation between pages | Menu/header |

# 7. Non-Functional Requirements

| ID | Requirement | Target |
|---|---|---|
| NFR-01 | File upload processing time | < 10 seconds |
| NFR-02 | Dashboard load time | < 3 seconds |
| NFR-03 | AI explanation generation | < 15 seconds |
| NFR-04 | Concurrent users supported | 10+ |
| NFR-05 | Data accuracy | 100% (no rounding errors) |
| NFR-06 | Browser support | Chrome, Firefox, Safari, Edge |
| NFR-07 | Mobile responsive | Yes |
| NFR-08 | Data persistence | SQLite (MVP), PostgreSQL (advanced) |
| NFR-09 | API response format | JSON |
| NFR-10 | Error handling | Graceful with user-friendly messages |

# 8. System Architecture




# 9. Database Schema

## 9.1 Table: companies

| Column | Type | Constraints | Description |
|---|---|---|---|
| id | INTEGER | PRIMARY KEY, AUTOINCREMENT | Unique ID |
| name | TEXT | NOT NULL | Company name |

| Column | Type | Constraints | Description |
|---|---|---|---|
| created_at | TIMESTAMP | DEFAULT CURRENT_TIMESTAMP | Upload date |
| updated_at | TIMESTAMP | DEFAULT CURRENT_TIMESTAMP | Last update |

## 9.2 Table: financial_data

| Column | Type | Constraints | Description |
|---|---|---|---|
| id | INTEGER | PRIMARY KEY, AUTOINCREMENT | Unique ID |
| company_id | INTEGER | FOREIGN KEY | References companies(id) |
| year | INTEGER | NOT NULL | Financial year |
| revenue | REAL | NOT NULL | Total revenue |
| cogs | REAL | NOT NULL | Cost of goods sold |
| operating_expenses | REAL | | Operating expenses |
| net_profit | REAL | NOT NULL | Net profit |
| receivables | REAL | NOT NULL | Accounts receivable |
| inventory | REAL | NOT NULL | Inventory |
| total_assets | REAL | NOT NULL | Total assets |
| total_debt | REAL | NOT NULL | Total debt |
| equity | REAL | NOT NULL | Shareholders' equity |
| operating_cash_flow | REAL | NOT NULL | Operating cash flow |
| capex | REAL | NOT NULL | Capital expenditure |
| depreciation | REAL | NOT NULL | Depreciation |
| related_party_transactions | REAL | DEFAULT 0 | RPT amount |
| interest_expense | REAL | DEFAULT 0 | Interest expense |
| current_assets | REAL | | Current assets |
| current_liabilities | REAL | | Current liabilities |

## 9.3 Table: ratios

| Column | Type | Constraints | Description |
|---|---|---|---|
| id | INTEGER | PRIMARY KEY, AUTOINCREMENT | Unique ID |
| company_id | INTEGER | FOREIGN KEY | References companies(id) |
| year | INTEGER | NOT NULL | Financial year |
| gross_margin | REAL | | Gross margin |
| operating_margin | REAL | | Operating margin |
| net_margin | REAL | | Net profit margin |
| roe | REAL | | Return on equity |

| Column | Type | Constraints | Description |
|---|---|---|---|
| roa | REAL | | Return on assets |
| roce | REAL | | Return on capital employed |
| current_ratio | REAL | | Current ratio |
| quick_ratio | REAL | | Quick ratio |
| ocf_ratio | REAL | | Operating cash flow ratio |
| debt_to_equity | REAL | | Debt-to-equity |
| debt_to_assets | REAL | | Debt-to-assets |
| interest_coverage | REAL | | Interest coverage |
| receivables_turnover | REAL | | Receivables turnover |
| inventory_turnover | REAL | | Inventory turnover |
| asset_turnover | REAL | | Asset turnover |
| cash_conversion_cycle | REAL | | Cash conversion cycle |
| ocf_to_profit | REAL | | OCF / Net Profit |
| free_cash_flow | REAL | | Free cash flow |

## 9.4 Table: anomalies

| Column | Type | Constraints | Description |
|---|---|---|---|
| id | INTEGER | PRIMARY KEY, AUTOINCREMENT | Unique ID |
| company_id | INTEGER | FOREIGN KEY | References companies(id) |
| anomaly_type | TEXT | NOT NULL | A, B, C, D, E, F, or G |
| year | INTEGER | NOT NULL | Year detected |
| description | TEXT | NOT NULL | What was found |
| risk_points | INTEGER | NOT NULL | Points added |
| created_at | TIMESTAMP | DEFAULT CURRENT_TIMESTAMP | Detection date |

## 9.5 Table: risk_scores

| Column | Type | Constraints | Description |
|---|---|---|---|
| id | INTEGER | PRIMARY KEY, AUTOINCREMENT | Unique ID |
| company_id | INTEGER | FOREIGN KEY | References companies(id) |
| score | INTEGER | NOT NULL | 0–100 |
| risk_level | TEXT | NOT NULL | Low/Moderate/High/Very High |
| created_at | TIMESTAMP | DEFAULT CURRENT_TIMESTAMP | Calculation date |

## 9.6 Table: explanations

| Column | Type | Constraints | Description |
|---|---|---|---|
| id | INTEGER | PRIMARY KEY, AUTOINCREMENT | Unique ID |
| company_id | INTEGER | FOREIGN KEY | References companies(id) |
| anomaly_id | INTEGER | FOREIGN KEY | References anomalies(id) |
| detected | TEXT | | What was found |
| why_it_matters | TEXT | | Why it matters |
| recommended_investigation | TEXT | | What to check next |
| created_at | TIMESTAMP | DEFAULT CURRENT_TIMESTAMP | Generation date |

# 10. API Specifications

## 10.1 Base URL

```
http://localhost:8000/api/v1
```

## 10.2 Endpoints

### POST /upload

**Description:** Upload Excel/CSV file with financial data.
**Request:**

```
Content-Type: multipart/form-data

file: <binary file>
company_name: <string>
```

**Response:**

```
{
  "success": true,
  "company_id": 1,
  "message": "File uploaded and processed successfully",
  "years_found": 5,
  "data_points": 85
}
```

**Error Response:**

```
{
  "success": false,
  "error": "Invalid file format. Please upload .xlsx, .xls, or .csv"
}
```

### GET /company/{company_id}

**Description:** Get company details.

```
{
  "id": 1,
  "name": "Example Industries Ltd.",
  "created_at": "2026-01-15T10:30:00Z",
  "years": [2021, 2022, 2023, 2024, 2025],
  "financial_data": [
    {
      "year": 2021,
      "revenue": 1000,
      "net_profit": 100,
      "operating_cash_flow": 120,
      "receivables": 150,
```

```
      "inventory": 200
    }
  ]
}
```


## GET /analyze/{company_id}

**Description:** Run full analysis and return results.


```
{
  "company_id": 1,
  "company_name": "Example Industries Ltd.",
  "risk_score": 73,
  "risk_level": "High",
  "anomalies": [
    {
      "type": "A",
      "name": "Receivables vs Revenue",
      "year": 2025,
      "description": "Receivables grew 48% vs Revenue 12%",
      "risk_points": 15
    },
    {
      "type": "B",
      "name": "Profit vs OCF",
      "year": 2025,
      "description": "Profit grew 31% but OCF declined 18%",
      "risk_points": 20
    }
  ],
  "ratios": {
    "2021": { "gross_margin": 0.40, "net_margin": 0.10 },
    "2022": { "gross_margin": 0.41, "net_margin": 0.11 }
  }
}
```


## GET /anomalies/{company_id}

**Description:** Get list of detected anomalies.


```
{
  "company_id": 1,
  "anomalies": [
    {
      "id": 1,
      "type": "A",
      "name": "Receivables vs Revenue",
      "year": 2025,
      "description": "Receivables grew 48% vs Revenue 12%",
      "risk_points": 15
    }
  ]
}
```


## GET /explanations/{company_id}

**Description:** Get AI-generated explanations.


```
{
  "company_id": 1,
  "explanations": [
    {
      "anomaly_id": 1,
      "detected": "Accounts receivable increased by 46% while revenue increased by only 11%.",
      "why_it_matters": "Receivables growing substantially faster than revenue can indicate slower collections or potentially aggressive revenue rec
```

```
      "recommended_investigation": "Examine customer receivable aging, credit terms, subsequent collections, and revenue recognition policies."
    }
  ]
}
```


### GET /companies

**Description:** List all companies.


```
{
  "companies": [
    {
      "id": 1,
      "name": "Example Industries Ltd.",
      "risk_score": 73,
      "risk_level": "High",
      "created_at": "2026-01-15T10:30:00Z"
    }
  ]
}
```


# 11. Excel Template Specification

## 11.1 Required Columns

| Column Name | Data Type | Required? | Example |
|---|---|---|---|
| Year | Integer | Yes | 2021 |
| Revenue | Float | Yes | 1000.00 |
| COGS | Float | Yes | 600.00 |
| Operating Expenses | Float | No | 200.00 |
| Net Profit | Float | Yes | 100.00 |
| Accounts Receivable | Float | Yes | 150.00 |
| Inventory | Float | Yes | 200.00 |
| Total Assets | Float | Yes | 2000.00 |
| Total Debt | Float | Yes | 800.00 |
| Equity | Float | Yes | 1200.00 |
| Operating Cash Flow | Float | Yes | 120.00 |
| Capex | Float | Yes | 100.00 |
| Depreciation | Float | Yes | 80.00 |
| Related-Party Transactions | Float | No | 10.00 |
| Interest Expense | Float | No | 40.00 |
| Current Assets | Float | No | 500.00 |
| Current Liabilities | Float | No | 300.00 |

## 11.2 Sample Template

| Year | Revenue | COGS | Net Profit | Receivables | Inventory | Total Assets | Total Debt | Equity | OCF | Capex | Depreciation | RPT | Interest |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2021 | 1000 | 600 | 100 | 150 | 200 | 2000 | 800 | 1200 | 120 | 100 | 80 | 10 | 40 |
| 2022 | 1100 | 650 | 110 | 180 | 250 | 2200 | 850 | 1350 | 115 | 120 | 85 | 15 | 42 |
| 2023 | 1200 | 700 | 120 | 260 | 350 | 2400 | 950 | 1450 | 100 | 150 | 90 | 25 | 45 |
| 2024 | 1300 | 750 | 130 | 380 | 500 | 2600 | 1100 | 1500 | 85 | 180 | 95 | 45 | 50 |
| 2025 | 1400 | 800 | 140 | 550 | 700 | 2800 | 1300 | 1550 | 70 | 200 | 100 | 70 | 55 |

# 12. Frontend Specifications

## 12.1 Pages

| Page | Route | Purpose |
|---|---|---|
| Home | `/` | Landing page with upload button |
| Upload | `/upload` | File upload with drag-and-drop |
| Dashboard | `/dashboard/{company_id}` | Main results page |
| Company List | `/companies` | List all analyzed companies |
| Company Detail | `/company/{company_id}` | Deep dive into one company |

## 12.2 Dashboard Layout

Financial Fraud Detector — Dashboard

```
<div class="dash-card">
  <strong>Company: Example Industries Ltd.</strong>
  <div class="dash-score">73 / 100</div>
  <div class="dash-level">Risk Level: HIGH</div>
</div>

<div class="dash-card">
  <strong>Detected Anomalies</strong>
  <table>
    <thead>
      <tr><th>#</th><th>Anomaly</th><th>Risk</th><th>Points</th></tr>
    </thead>
    <tbody>
      <tr><td>1</td><td>Receivables Growth</td><td><span class="badge badge-high">High</span></td><td>+15</td></tr>
      <tr><td>2</td><td>Cash Flow Divergence</td><td><span class="badge badge-high">High</span></td><td>+20</td></tr>
      <tr><td>3</td><td>Inventory Growth</td><td><span class="badge badge-medium">Medium</span></td><td>+10</td></tr>
      <tr><td>4</td><td>Margin Anomaly</td><td><span class="badge badge-medium">Medium</span></td><td>+8</td></tr>
      <tr><td>5</td><td>Related-Party Transactions</td><td><span class="badge badge-medium">Medium</span></td><td>+12</td></tr>
      <tr><td>6</td><td>Depreciation vs Capex</td><td><span class="badge badge-medium">Medium</span></td><td>+10</td></tr>
      <tr><td>7</td><td>Debt &amp;amp; Interest</td><td><span class="badge badge-low">Low</span></td><td>+5</td></tr>
    </tbody>
  </table>
</div>

<div class="dash-card">
  <strong>Financial Trends</strong>
  <p>[Line Chart: Revenue, Net Profit, OCF over 5 years]</p>
  <p>[Bar Chart: Receivables, Inventory, Debt over 5 years]</p>
  <p>[Line Chart: Margins over 5 years]</p>
</div>

<div class="dash-card">
  <strong>AI Explanation — Anomaly #1: Receivables Growth</strong>
```

```
    <p><strong>Detected:</strong> Accounts receivable increased by 46% while revenue increased by only 11%.</p>
    <p><strong>Why it matters:</strong> Receivables growing substantially faster than revenue can indicate slower collections or aggressive revenue
    <p><strong>Recommended Investigation:</strong> Examine customer receivable aging, credit terms, and revenue recognition policies.</p>
  </div>

</div>
```

## 12.3 Components List

| Component | File | Props | Description |
|---|---|---|---|
| Header | Header.tsx | companyName | Top navigation bar |
| RiskScoreCard | RiskScoreCard.tsx | score, level | Big score display |
| AnomalyList | AnomalyList.tsx | anomalies[] | List of anomalies |
| AnomalyCard | AnomalyCard.tsx | anomaly | Single anomaly |
| TrendChart | TrendChart.tsx | data, lines[] | Line chart |
| BarChart | BarChart.tsx | data, bars[] | Bar chart |
| ExplanationCard | ExplanationCard.tsx | explanation | AI explanation |
| FileUpload | FileUpload.tsx | onUpload | Drag-drop upload |
| LoadingSpinner | LoadingSpinner.tsx | — | Loading indicator |
| ErrorMessage | ErrorMessage.tsx | message | Error display |

## 12.4 Colors & Styling

| Element | Color | Tailwind Class |
|---|---|---|
| Low Risk | Green | `bg-green-500` |
| Moderate-Low Risk | Yellow | `bg-yellow-500` |
| Moderate Risk | Orange | `bg-orange-500` |
| High Risk | Red | `bg-red-500` |
| Very High Risk | Dark Red | `bg-red-700` |
| Background | Light Gray | `bg-gray-100` |
| Card Background | White | `bg-white` |
| Text Primary | Dark Gray | `text-gray-900` |
| Text Secondary | Medium Gray | `text-gray-600` |

# 13. Backend Specifications

## 13.1 Project Structure

```
backend/
├── app/
│   ├── __init__.py
│   ├── main.py                # FastAPI app entry
│   ├── config.py              # Configuration
│   ├── database.py            # Database connection
│   ├── models/
│   │   ├── __init__.py
│   │   ├── company.py         # Company model
```

```
│   │   ├── financial_data.py   # Financial data model
│   │   ├── anomaly.py          # Anomaly model
│   │   └── risk_score.py       # Risk score model
│   ├── schemas/
│   │   ├── __init__.py
│   │   ├── company.py          # Pydantic schemas
│   │   ├── anomaly.py
│   │   └── risk_score.py
│   ├── services/
│   │   ├── __init__.py
│   │   ├── extraction.py       # Data extraction
│   │   ├── ratios.py           # Ratio calculation
│   │   ├── anomaly.py          # Anomaly detection
│   │   ├── scoring.py          # Risk scoring
│   │   └── ai_explanation.py   # AI explanation
│   ├── routers/
│   │   ├── __init__.py
│   │   ├── upload.py           # Upload endpoint
│   │   ├── analyze.py          # Analyze endpoint
│   │   └── company.py          # Company endpoints
│   └── utils/
│       ├── __init__.py
│       └── validators.py       # File validation
├── requirements.txt
└── .env
```

### 13.2 Dependencies

```
fastapi==0.104.1
uvicorn==0.24.0
pandas==2.1.3
numpy==1.26.2
scipy==1.11.4
scikit-learn==1.3.2
openpyxl==3.1.2
python-multipart==0.0.6
sqlalchemy==2.0.23
pydantic==2.5.2
openai==1.3.5
python-dotenv==1.0.0
```

## 14. Frontend Project Structure

```
frontend/
├── app/
│   ├── layout.tsx              # Root layout
│   ├── page.tsx                # Home page
│   ├── upload/
│   │   └── page.tsx            # Upload page
│   ├── dashboard/
│   │   └── [companyId]/
│   │       └── page.tsx        # Dashboard page
│   ├── companies/
│   │   └── page.tsx            # Company list
│   └── company/
│       └── [companyId]/
│           └── page.tsx        # Company detail
├── components/
│   ├── Header.tsx
│   ├── RiskScoreCard.tsx
│   ├── AnomalyList.tsx
│   ├── AnomalyCard.tsx
│   ├── TrendChart.tsx
│   ├── BarChart.tsx
│   ├── ExplanationCard.tsx
```

```
│   ├── FileUpload.tsx
│   ├── LoadingSpinner.tsx
│   └── ErrorMessage.tsx
├── lib/
│   ├── api.ts                # API calls
│   └── types.ts              # TypeScript types
├── public/
│   └── logo.png
├── package.json
├── tailwind.config.js
├── next.config.js
└── tsconfig.json
```

### 14.1 Frontend Dependencies

```
{
  "dependencies": {
    "next": "14.0.3",
    "react": "18.2.0",
    "react-dom": "18.2.0",
    "recharts": "2.10.3",
    "axios": "1.6.2",
    "tailwindcss": "3.3.6",
    "lucide-react": "0.294.0"
  },
  "devDependencies": {
    "typescript": "5.3.2",
    "@types/react": "18.2.42",
    "@types/node": "20.10.3"
  }
}
```

## 15. Error Handling

### 15.1 Error Types

| Error Code | Message | HTTP Status | When |
|---|---|---|---|
| `INVALID_FILE_FORMAT` | Invalid file format. Please upload .xlsx, .xls, or .csv | 400 | Wrong file type |
| `FILE_TOO_LARGE` | File size exceeds 10MB limit | 400 | File > 10MB |
| `MISSING_COLUMNS` | Missing required columns: {columns} | 400 | Required columns absent |
| `INSUFFICIENT_DATA` | At least 3 years of data required | 400 | < 3 years |
| `COMPANY_NOT_FOUND` | Company not found | 404 | Invalid company ID |
| `ANALYSIS_FAILED` | Analysis failed. Please try again | 500 | Internal error |
| `AI_SERVICE_ERROR` | AI explanation service unavailable | 503 | LLM API down |

### 15.2 Edge Cases

| Case | Handling |
|---|---|
| Only 1 year of data | Show warning, skip year-over-year analysis |
| Only 2 years of data | Run analysis, show warning about limited data |
| Missing optional columns | Use defaults (0) |
| Null values in required columns | Skip that row, show warning |
| Division by zero | Handle gracefully (return null or 0) |

| Case | Handling |
|---|---|
| Negative values | Allow (companies can have losses) |
| Very large numbers | Handle with proper formatting |

# 16. Acceptance Criteria

## 16.1 Upload

User can upload .xlsx, .xls, .csv files
File size limit of 10MB enforced
Invalid format shows error message
Missing columns show error message
Successful upload shows success message
Company ID returned after upload

## 16.2 Analysis

All 18 ratios calculated correctly
All 7 anomaly rules checked
Risk points assigned correctly
Total score capped at 100
Risk level assigned correctly

## 16.3 Dashboard

Risk score displayed prominently
Risk level color-coded
All detected anomalies listed
Risk points shown for each anomaly
Charts render correctly
AI explanations displayed

## 16.4 AI Explanation

Explanation generated for each anomaly
Has 3 parts (Detected, Why, Investigation)
Language is simple
No fraud claims made
Stored in database

## 16.5 Performance

Upload processed within 10 seconds
Dashboard loads within 3 seconds
AI explanation generated within 15 seconds
Supports 10+ concurrent users

# 17. MVP Scope vs Advanced Scope

## 17.1 MVP (Minimum Viable Product)

| Feature | Included? |
|---|---|
| Excel/CSV upload | ✅ Yes |
| 3–5 years of data | ✅ Yes |
| Ratio calculation (18 ratios) | ✅ Yes |
| 7 anomaly detection rules | ✅ Yes |
| Risk score (0–100) | ✅ Yes |
| Interactive charts | ✅ Yes |
| AI explanations | ✅ Yes |
| SQLite database | ✅ Yes |

| Feature | Included? |
|---|---|
| Basic dashboard | ✅ Yes |

### 17.2 Advanced (Future)

| Feature | Included? |
|---|---|
| PDF extraction | ❌ No |
| OCR | ❌ No |
| Notes-to-accounts analysis | ❌ No |
| Related-party transaction extraction from notes | ❌ No |
| Machine-learning anomaly detection | ❌ No |
| Peer comparison | ❌ No |
| AI-generated investigation questions | ❌ No |
| Automated financial analysis report | ❌ No |
| PostgreSQL database | ❌ No |
| Multi-user authentication | ❌ No |
| Export to PDF/Excel | ❌ No |
| Anomaly timeline visualization | ❌ No |
| Industry benchmarking | ❌ No |

## 18. Limitations

**Not proof of fraud:** High risk score means "deserves investigation," not "committed fraud."
**Historical data only:** System uses past data, not real-time.
**No industry comparison:** MVP doesn't compare with peers.
**Limited to 7 anomalies:** More fraud patterns exist.
**Requires clean data:** Manual Excel/CSV input needed for MVP.
**AI explanations are suggestions:** Not definitive conclusions.

## 19. Success Metrics

| Metric | Target |
|---|---|
| Upload success rate | > 95% |
| Analysis accuracy | 100% (deterministic) |
| Dashboard load time | < 3 seconds |
| AI explanation quality | > 4/5 user rating |
| User can complete analysis in | < 2 minutes |
| System uptime | > 99% |

## 20. Development Timeline

| Week | Task | Deliverable |
|---|---|---|
| 1 | Setup project structure | Backend + frontend scaffolding |

| Week | Task | Deliverable |
|---|---|---|
| 2 | Data layer | Excel template, database, extraction |
| 3 | Ratio engine | All 18 ratios calculated |
| 4 | Anomaly engine | All 7 rules implemented |
| 5 | Risk scoring | Score calculation + risk levels |
| 6 | Backend API | All endpoints working |
| 7 | Frontend - Upload | Upload page functional |
| 8 | Frontend - Dashboard | Dashboard with score + anomalies |
| 9 | Frontend - Charts | All charts rendering |
| 10 | AI integration | Explanations generated |
| 11 | Testing | All acceptance criteria met |
| 12 | Documentation | README + user guide |

# 21. Glossary

| Term | Definition |
|---|---|
| **Anomaly** | Unusual pattern that may indicate a problem |
| **COGS** | Cost of Goods Sold |
| **Capex** | Capital Expenditure |
| **OCF** | Operating Cash Flow |
| **RPT** | Related-Party Transactions |
| **ROE** | Return on Equity |
| **ROA** | Return on Assets |
| **ROCE** | Return on Capital Employed |
| **PRD** | Product Requirements Document |
| **MVP** | Minimum Viable Product |
| **LLM** | Large Language Model |
| **API** | Application Programming Interface |

# 22. Appendix: Sample AI Explanations

## Anomaly A: Receivables vs Revenue

**Detected:** Accounts receivable increased by 46% while revenue increased by only 11%.
**Why it matters:** Receivables growing substantially faster than revenue can indicate slower collections or potentially aggressive revenue recognition. When a company books sales faster than it collects cash, it may be recognizing revenue before it's truly earned.
**Recommended Investigation:**
Examine customer receivable aging
Review credit terms offered to customers
Check subsequent collections after year-end
Review revenue recognition policies

## Anomaly B: Profit vs OCF

**Detected:** Net profit increased by 31% but operating cash flow decreased by 18%.
**Why it matters:** When accounting profit rises but actual cash generation falls, it may indicate poor earnings quality. Profit is an accounting opinion; cash is a fact. Divergence between the two is a classic warning sign of potential earnings manipulation.
**Recommended Investigation:**
Review accrual accounts (receivables, payables, inventory)
Check for one-time gains inflating profit
Examine revenue recognition timing
Compare cash flow from operations to net income over multiple years

# 23. Document Approval

| Role | Name | Signature | Date |
|---|---|---|---|
| Product Owner | | | |
| Technical Lead | | | |
| Finance Domain Expert | | | |
| Project Manager | | | |

**Financial Fraud Detector — Product Requirements Document**
Document Version: 1.0 | Last Updated: 2026 | Next Review: After MVP completion
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